

Bharat Steel & Alloys Ltd.

BSE / NSE: BSAL · Steel — Integrated Producer · India · FY2024

CLIMATE INTELLIGENCE SCORE	CONFIDENCE	PEER RANKING
B 40.3 / 100	HIGH Score: 78 / 100	28th Percentile · Steel-IN-Large

OUTLOOK	STABLE / POSITIVE POTENTIAL	ASSESSMENT YEAR	FY2024	NEXT REVIEW	June 2027
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REPORT TYPE	Full-Spectrum Climate Intelligence Assessment	ENGINE VERSION	Climactix Intelligence Engine v1.0
ASSESSMENT ID	CX-2026-BSAL-001	METHODOLOGY REF	ARCHITECTURE.md v1.0
PREPARED BY	Climactix Global Intelligence Engine	CLASSIFICATION	Institutional — Authorized Recipients Only

This report is generated from structured assessment data submitted by Bharat Steel & Alloys Ltd. and publicly available climate science databases. Financial projections are model-based estimates. This report does not constitute investment advice. Scores reflect climate risk management quality — not creditworthiness or equity value. For authorized recipients only.

EXECUTIVE SUMMARY

Bharat Steel & Alloys Ltd.

Climate Intelligence Score: **B (40.3 / 100)** · 28th Percentile · Confidence: HIGH · Outlook: Stable / Positive Potential

BSAL is a company in transition — one that discloses consistently, governs adequately, and has begun reducing emissions intensity — but has not yet built the institutional climate risk infrastructure required by capital market standards now being set by SEBI, RBI, and global institutional investors.

THREE STRUCTURAL VULNERABILITIES

01 Carbon Lock-in

BF-BOF technology with 15–20 years remaining asset life. Scope 1 intensity 1.92 tCO₂e/t — above the sector median (1.85). No technology transition investment visible in disclosed CAPEX. Carbon pricing will consume an estimated 59% of FY2024 EBITDA by 2030 under Disorderly pricing.

02 Climate Intelligence Deficit

No scenario analysis conducted or disclosed. No physical risk assessment completed. Net-Zero 2050 aspiration without roadmap, interim milestones, or capital allocation. Climate Resilience score: 13.8/100 — bottom decile of peer cohort.

03 Supply Chain Opacity

12% of Tier 1 suppliers assessed for climate or ESG risk. 78% of coking coal sourced from a single country (Australia). No formal supplier ESG questionnaire programme. Supply Chain score: 19.3/100 — 12th percentile of peer cohort.

For institutional investors: BSAL is a borderline hold at current climate risk management maturity. Trajectory is positive but speed of progress is insufficient given the 2030 financial exposure horizon. Recommend conditional engagement with a 12-month reassessment trigger tied to SBTi submission and scenario analysis completion.

CLIMATE INTELLIGENCE SCORECARD

Full Domain Assessment

OVERALL CIS	B	40.3 / 100	28th Percentile Steel–India–Large	HIGH Confidence	STABLE / POS. POTENTIAL
GOVERNANCE		60	BBB	55th pct	PROGRESSING
PHYSICAL RISK		50	BB+	47th pct	DEVELOPING
TRANSITION RISK		28	CCC	16th pct	LAGGARD
CARBON MANAGEMENT		33	CCC	18th pct	LAGGARD
SUPPLY CHAIN		19	D	12th pct	LAGGARD
COMPLIANCE		50	BB	48th pct	DEVELOPING
ADAPTATION		47	BB	52th pct	PROGRESSING
DISCLOSURE QUALITY		74	A	78th pct	ADVANCED
GREENWASHING RISK		42	—	60th pct	ELEVATED
CLIMATE RESILIENCE		14	D	8th pct	LAGGARD

The B rating reflects a company with functional governance and disclosure infrastructure but without climate risk management depth. The wide dispersion — Disclosure Quality at A (78th percentile) alongside Carbon Management at CCC (18th) — is characteristic of strong form without substance.

01. *Ambition & Target*

Emissions baseline, trajectories and target credibility

01 — AMBITION & TARGETS

Our emissions baseline

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Our emissions baseline

Recognising where emissions come from is the foundation of effective climate strategy. BSAL's FY2024 baseline confirms that Scope 1 direct emissions — primarily from the Blast Furnace–Basic Oxygen Furnace steelmaking route — represent the dominant and most financially material emission source.

BSAL's Scope 1 intensity of **1.92 tCO₂e per tonne of crude steel** is 3.8% above the Steel-India-Large cohort median of 1.85 tCO₂e/t. This above-median position drives a zero score on the emissions performance sub-component — the largest single weight (0.35) in the Carbon Management domain.

The three-year emissions intensity trend (2.94%/yr decline) is broadly Paris-aligned for the steel sector. If sustained, BSAL crosses below the sector median by FY2027 — at which point the Carbon Management score moves from 33 to approximately 68.

EMISSIONS PROFILE — FY2024

Scope	Absolute	Intensity
Scope 1 (Direct)	5,376,000 t	1.92 tCO ₂ e/t
Scope 2 Market-based	892,000 t	0.32 tCO ₂ e/t
Scope 3 (Partial)	Estimated	Cat. 1, 11 only
Sector Median (p50)	—	1.85 tCO ₂ e/t

3-YEAR INTENSITY TREND

Year	Scope 1 Intensity	YoY Change
FY2022	2.04 tCO ₂ e/t	—
FY2023	1.98 tCO ₂ e/t	-2.9%
FY2024	1.92 tCO ₂ e/t	-3.0%

2.94% per year

TARGET CREDIBILITY ASSESSMENT

Net-Zero 2050: aspiration without architecture

BSAL has declared a Net-Zero by 2050 target in its Annual Report and Sustainability Policy. The Climactix Target Credibility engine assessed this commitment against five institutional criteria — and found material gaps across all five.

Criterion	Status	Score
SBTi Validation	Not submitted	0 / 25
Interim 2030 milestone	Not defined	0 / 25
CAPEX allocation to decarbonisation	Undisclosed	0 / 20
Technology transition roadmap	Not published	0 / 20

Criterion	Status	Score
Board approval of target	Yes — Sustainability Policy	30 / 10
A Net-Zero 2050 commitment from a BF-BOF steel company without a technology transition plan, without SBTi validation, and without disclosed CAPEX allocation is not an institutional-grade target. Target Credibility Score: 30 / 100.		

02. *Climate Risk Exposure*

Physical hazards, asset vulnerability and adaptation

02 — CLIMATE RISK EXPOSURE

Physical risk and adaptation

BSAL operates across two geographically distinct risk environments. The Raipur primary site (3.2 MTPA) faces inland water stress and heat risk. The Vizag rolling mill (0.8 MTPA) carries coastal hazard exposure. Together, they create a compound physical risk profile that scores BB+ (49.5) — 47th percentile of peers.

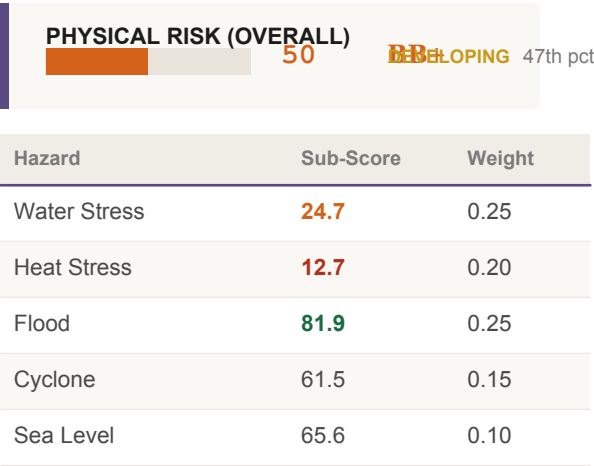
Hazard	Site	Level	Detail	2050 Trend
Water Stress	Raipur — Primary	CRITICAL	WRI score: 3.8/5 · 62% capacity in high-stress zones	Worsening
Heat Stress	Raipur	HIGH	WBGT p95: 29.8°C · Heavy work threshold	Worsening
Cyclone	Vizag — Coastal	HIGH	Bay of Bengal · 18% assets coastal	Intensifying
Flood	Vizag	MEDIUM	9% assets in 100-yr flood zone	Worsening
Sea Level Rise	Vizag	MEDIUM	4.2 km coastal · IPCC SLR trajectory	Accelerating

WATER STRESS — CRITICAL EXPOSURE

This is BSAL's most pressing physical risk. 62% of manufacturing capacity is in WRI High or Extremely High water stress zones. Integrated steel manufacturing requires water for blast furnace cooling, coke quenching, and continuous casting — the process is not discretionary.

BSAL's Zero Liquid Discharge (ZLD) system at Raipur provides genuine mitigation — reducing fresh water consumption through recirculation, with CPCB certificate verified. However, ZLD addresses liquid discharge, not raw water procurement risk. No formal water risk assessment or stewardship strategy was identified in any uploaded document.

PHYSICAL RISK SCORE BREAKDOWN



03. *Transition Readiness*

Carbon pricing, energy efficiency and technology lock-in

03 — TRANSITION READINESS

Transition Risk: the most material near-term financial exposure

Critical finding: Carbon pricing alone will consume an estimated **59% of FY2024 EBITDA by 2030** under Disorderly Transition pricing (\$15/tCO₂e). At 2040 prices (\$60/tCO₂e), the exposure exceeds total current EBITDA. Transition Risk domain score: **CCC (28.2 / 100)** — 16th percentile of peers.

CARBON PRICING COST PROJECTION — DISORDERLY SCENARIO

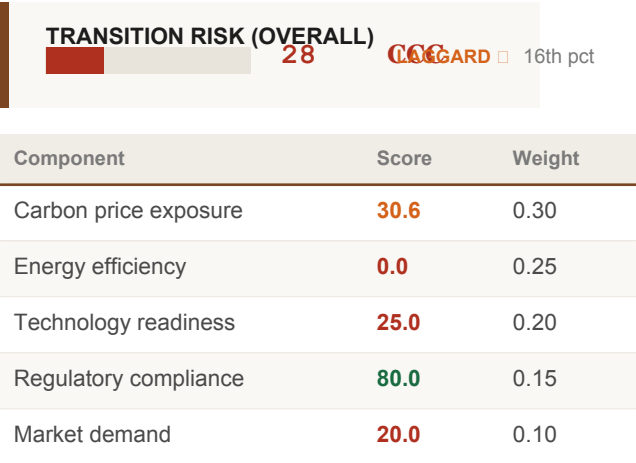
Metric	Now (FY26)	2030	2040	2050
Carbon Price (\$/tCO ₂ e)	\$8–12	\$15–20	\$60–80	\$150–180
BSAL Scope 1 (with trajectory)	5,376,000 t	4,868,000 t	3,782,000 t	2,937,000 t
Annual Carbon Cost	\$53M	\$73M	\$227M	\$441M
As % of FY24 EBITDA (\$124M)	42.7%	58.9%	183%	356%

TECHNOLOGY LOCK-IN

BSAL's BF-BOF equipment at Raipur was refurbished in 2019, implying remaining asset life to approximately 2039–2044. No investment in DRI-H2 or EAF technology has been disclosed. Each year of continued BF-BOF investment without a transition plan represents additional stranded asset accumulation.

BSAL's energy intensity of 23.4 GJ/t is 18.2% above the sector median (19.8 GJ/t). Annual energy cost is approximately \$205M — 33% of revenue. This drives a zero score on the energy efficiency sub-component, the second-largest transition weight (0.25).

TRANSITION SCORE BREAKDOWN



FINANCIAL RISK SIMULATION

Climate risk in financial terms

Methodology: NGFS 2023 scenarios × IPCC AR6. Base year: FY2024 (Revenue \$620M, EBITDA \$124M). Uncertainty band: ±35% at 80% confidence.

Risk Type	Scenario	2030 (% Rev)	2040 (% Rev)	2050 (% Rev)
Physical Risk	Orderly NZ2050	\$11.1M (1.8%)	\$24.1M (3.9%)	\$41.6M (6.7%)
Physical Risk	Disorderly ~3°C	\$19.7M (3.2%)	\$44.7M (7.2%)	\$92.4M (14.9%)
Physical Risk	Hot House ~4°C	\$28.1M (4.5%)	\$68.5M (11.1%)	\$151.8M (24.5%)
Transition Risk	Orderly NZ2050	\$82.7M (13.3%)	\$193M (31.2%)	\$321M (51.8%)
Transition Risk	Disorderly ~3°C	\$128.5M (20.7%)	\$406M (65.5%)	\$819M (132%)
Transition Risk	Hot House ~4°C	\$31.4M (5.1%)	\$45.9M (7.4%)	\$71.9M (11.6%)

Key insight: The most financially dangerous scenario for BSAL is **Disorderly Transition** — not Hot House. Under Hot House, carbon prices remain low. Under Disorderly, prices rise faster than the company can adapt while physical risk simultaneously intensifies. The 132% revenue exposure by 2050 represents an existential scenario without material decarbonisation investment by 2028.

04. *Governance & Disclosure*

Board oversight, accountability and framework readiness

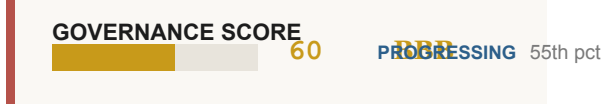
04 — GOVERNANCE & DISCLOSURE

Governance: strong form, incomplete substance

BSAL's governance structure is substantively ahead of many Steel sector peers. The Environment and Sustainability Committee, established at Board level in FY2022, has verified Terms of Reference. A board-approved Sustainability Policy (FY2023) provides the foundational governance document.

MATERIAL GAP

No third-party assurance exists on any sustainability data. BSAL's emissions, energy, and water data is entirely self-reported. There is no ISAE 3000 verification, no ISO 14064 audit, and no specialist firm review. This is the single highest-leverage governance action available — commissioning ISAE 3000 limited assurance would alone add 8–10 points to Governance and cascade to Confidence Score.



SCORE BREAKDOWN

Component	Score	Weight
Board oversight	100.0	0.30
ESG policy quality	65.0	0.25
Executive accountability	40.0	0.20
Audit / verification	0.0	0.15
Stakeholder engagement	55.0	0.10

DISCLOSURE FRAMEWORK READINESS

Framework	Readiness	Status	Assessment
BRSR Core	78%	Required — Top Listed	Near-ready
BRSR Full	58%	Required	Gap work needed
TCFD	36%	Required (upcoming)	Major gaps — Scenario analysis
IFRS S2	25%	Required (2-3 yrs)	Critical gaps
GRI	59%	Voluntary	Reasonable
CDP Climate	40%	Voluntary	Medium gaps

05. *Priority Action Plan*

Ranked by combined score impact and financial risk reduction

PRIORITY ACTION PLAN

What needs to change

TIER 1 — CRITICAL (within 6 months)**01 Conduct formal climate scenario analysis**

Resolves TCFD Strategy.b and IFRS S2 Para 22 gaps. Raises Climate Resilience from 13.8 to ~53 (+39.2 pts). CIS impact: +2.0 points.

3–4 months · □30–60 Lakhs · CFO + CSO + Board

02 Submit SBTi Commitment Letter

Target credibility moves 30 → 65. Carbon Management rises 33.0 → ~41.3. CIS impact: +1.5 points. Even commitment (unvalidated) moves score.

1 month · Low cost · Chief Sustainability Officer

03 Commission third-party ESG data assurance (ISAE 3000)

Confidence Score moves 78 → ~86 (VERY HIGH). Unlocks credit-grade institutional investor access. Verification Status 55 → 90.

2–3 months · □20–40 Lakhs · CFO + External Auditor

TIER 2 — HIGH PRIORITY (within 12 months)**04 Publish standalone carbon inventory (ISO 14064)**

Data quality sub-score in Carbon Management 60 → 85. CMS gains ~5 points. Resolves Scope 3 evidence gap.

3–4 months · □15–25 Lakhs

05 Launch formal Supplier ESG Audit Programme

Supply Chain score rises from 19.3 toward 35+ over 2–3 years. Target: 40% Tier 1 coverage by FY2026. Start with top 20 suppliers by spend.

6–9 months ongoing · □10–20 Lakhs

06 Commission physical risk assessment — both sites

Physical Risk score rises to ~60. Closes TCFD Risk Management.a gap. Enables IFRS S2 Para 9 disclosure.

3–4 months · □25–40 Lakhs

07 Expand executive climate KPIs across full C-suite

Executive accountability 40 → 75. Governance rises from 59.8 to ~65. Introduce Scope 1 intensity, water recycling, supplier audit coverage.

1–2 months · Board NRC approval

RATING RATIONALE & INSTITUTIONAL GUIDANCE

Rating: B (40.3) · Outlook: Stable / Positive Potential

POSITIVE FACTORS

- + Consistent BRSR Core disclosure — evidence-verified
- + Board ESG committee with documented mandate (FY2022)
- + Paris-aligned Scope 1 intensity trajectory: -2.94%/yr
- + ZLD system at Raipur — verified CPCB certificate
- + PAT Scheme Cycle 5 target achieved
- + Disclosure Quality: A rating, 78th peer percentile
- + HIGH confidence (78) — current, near-complete data

NEGATIVE FACTORS

- Scope 1 intensity above sector median → zero performance score
- No scenario analysis of any kind conducted or disclosed
- Net-Zero 2050 claim with no roadmap or CAPEX plan
- Supply chain: 12% supplier coverage, bottom decile
- Energy intensity 18.2% above sector median
- Climate Resilience: D, 8th percentile of peers
- No third-party ESG data assurance

UPGRADE TRIGGERS — TO BB OR BB+

- › SBTi commitment submitted (achievable in < 3 months)
- › Third-party ISAE 3000 assurance on FY2025 sustainability data
- › Scope 1 intensity falls below 1.85 tCO₂e/t (on trajectory: FY2027)
- › Formal climate scenario analysis published (TCFD-aligned)

INSTITUTIONAL INVESTOR GUIDANCE

Use Case	Suitability	Rationale
ESG / Climate equity fund	CONDITIONAL HOLD	Monitor SBTi submission. Below Paris-aligned threshold for most mandates.
Investment grade bond / NCD	REQUIRES IMPROVEMENT	Commission ESG assurance before credit mandate. Confidence HIGH but not VERY HIGH.
Green / SLL bond	NOT ELIGIBLE (current)	No SBTi target. No verified KPI framework. Eligible after SBTi + scenario analysis.
Infrastructure / project finance	PROCEED WITH CONDITIONS	Physical risk assessment required as loan condition. Carbon stress-test in project model.
Pension fund (PA mandate)	BELOW THRESHOLD	B rating (28th pct) below typical PA mandate minimums (BB+ / A). Reassess in 12 months.

CIS RATING
B (40.3)

CONFIDENCE
HIGH (78)

OUTLOOK
STABLE /
POS.

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